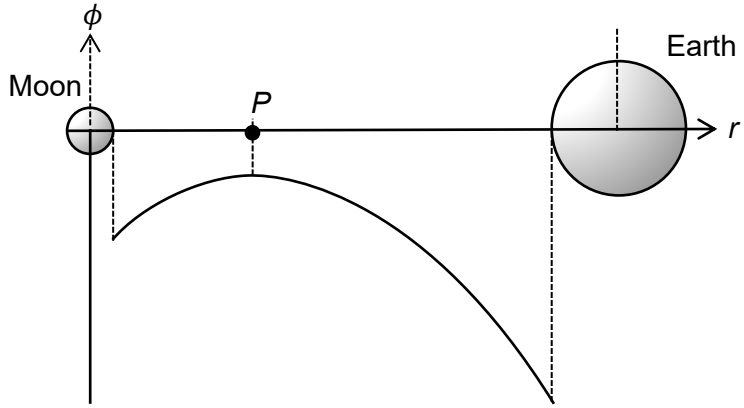
10 The gravitational potential ϕ as it varies with distance r from the centre of the Moon to the centre of Earth as shown in the diagram below.



Which of the following statements is **not** true?

A The gravitational field strength at *P* is zero.

B The gravitational field strength at *P* is the scalar sum of the individual field strengths of the Moon and the Earth at point *P*.

C The gravitational field strength at *P* is the vector sum of the individual field strengths of the Moon and the Earth at point *P*.

D The gravitational field strength at *P* is given by the rate of change of ϕ with respect to *r*.